II Year B.Tech. ME – II Semester

L	T	P	C
2	0	0	2

(23A52402d) INDUSTRIAL MANAGEMENT

Course Objectives: The objectives of the course are to

- Introducethescopeandroleofindustrialengineeringandthetechniquesforoptimaldesignofl ayouts
- Illustrate how work study is used to improve productivity
- Explain TQM and quality control techniques
- Introduce financial management aspects and
- Discuss human resource management and value analysis.

Course Outcomes:

COs	Statements	BloomsLevel
CO1	Learn about how to design the optimal layout	L1
CO2	Demonstrate work study methods	L3
CO3	Explain Quality Control techniques	L2
CO4	Discuss the financial management aspects	L3
CO5	Understand the human resource management methods.	L2

UNIT– I

INTRODUCTION: Definition of industrial engineering (I.E), development, applications, role of anindustrial engineer, quantitativetoolsofIEandproductivitymeasurement.Conceptsofmanagement,importance,functionsofman agement, scientific management, Taylor's principles, Fayol's principles ofmanagement.
PLANTLAYOUT: Factorsgoverningplantlocation,typesofproductionlayouts,advantagesanddisadvantag esofprocesslayoutandproductlayout,applications,quantitativetechniquesforoptimaldesignof layouts.

UNIT–II

WORK STUDY: Importance, types of production, applications, work study, method study and timestudy,worksampling,PMTS,micro-motionstudy,ratingtechniques,MTM,workfactorsystem,principlesofErgonomics,flowprocesscharts,strin gdiagramsandTherbligs.

UNIT–III

STATISTICAL QUALITY CONTROL: Quality control, Queuing assurance and its importance, SQC,attribute sampling inspection with single and double sampling, Control charts – X and R –charts X and Scharts and their applications, simple numerical examples.
TOTALQUALITYMANAGEMENT: Elements of TQM – Continuous Improvement –

zerodefectconcept,qualitycircles,implementation,applications,ISOqualitysystems.SixSigma–definition,basic concepts.

UNIT– IV

FINANCIAL MANAGEMENT: Scope and nature of financial management, Sources of finance, Management of working capital, estimation of working capital requirements, budget and budgetary control, Capital budgeting – Nature of InvestmentDecisions–InvestmentEvaluationcriteria-NPV,IRR,PI,PaybackPeriod,andARR,numericalproblems.

UNIT–V

HUMAN RESOURCE MANAGEMENT: Concept of human resource management, personnel management and industrial relations, functions of personnel management, Job-evaluation, its importance and types, merit rating, quantitative methods, wage incentive plans, and types.
VALUE ANALYSIS: Value engineering, implementation procedure, enterprise resource planning and supply chain management.

Text Books:

1. O.PKhanna, Industrial Engineering and Management, Dhanpat Rai Publications (P)Ltd.
2. Mart and Telsang, Industrial Engineering and Production Management, S.Chand & Company Ltd.New Delhi

Reference Books:

1. Bhattacharya DK, Industrial Management, S.Chand, publishers.
2. J.G Monks, Operations Management, 3/e, McGraw Hill Publishers.
3. T.R.Banga,S.C.Sharma,N.K.Agarwal,IndustrialEngineeringandManagementScience,KhannaPublishers.
4. Koontz O'Donnell, Principles of Management, McGraw Hill Publishers.
5. R.C.Gupta, Statistical Quality Control, Khanna Publishers.
6. NVS Raju, Industrial Engineering and Management, Cengage India Private Limited.

Online Learning Sources

- https://onlinecourses.nptel.ac.in/noc21_me15/preview
- https://onlinecourses.nptel.ac.in/noc20_mg43/preview
- https://www.edx.org/learn/industrial-engineering
- https://youtube.com/playlist?list=PL299B5CC87110A6E7&si=TghLCbEobuxjEaXi
- https://youtube.com/playlist?list=PLbjTnj-t5Gkl0z3OHOGK5RB9mvNYvnImW&si=oaX_5RG69hS3v2ll